

```
import pandas as pd
import numpy as np

campaign = pd.read_excel('CampaignSubjects.xlsx')
choice = pd.read_excel('ChoiceSubjectsData.xlsx')
```

```
campaign
```

	Id	SubjectType \
0	"f36cb6ff-8df4-4842-b59f-d05153bbb560"	SingleSelect
1	"d7f7cebb-baa8-41ce-95c5-688f86e50a75"	SingleSelect
2	"80011846-c284-4aa7-9742-8c3f16e131b7"	SingleSelect
3	"3c327491-9c15-4ab3-bdb9-8d2aafbbf87e"	SingleSelect
4	"e0b39b39-682d-462d-ab23-8de2577628fe"	LocationData
5	"9afaec0d-f7a8-4cc7-90a2-b77d5d62c922"	SingleSelect
6	"ecb57a0f-3779-4ded-99a7-1fa4d9c6a14f"	SingleSelect
7	"90c36fb3-1cf8-4659-b2bd-6d1dce401ba0"	LocationData
8	"8e630a29-7f74-41f8-9023-813459d7e254"	SingleSelect
9	"90f8fee2-9939-4dc3-a91c-2da82a2f8053"	SingleSelect
10	"e4eeb58a-5c80-428d-bf24-7372867dd28d"	MutliSelect
11	"935791bc-4831-4386-aa99-90f3c3ddb571"	SingleSelect
12	"044a1964-ec46-4404-9089-120f41dd0f54"	SingleSelect
13	"e8e6cdbe-9500-4eef-a117-1a31c5c35990"	SingleSelect
14	"8ca14928-2fbe-42b6-8713-5b4e17459f80"	SingleSelect
15	"a3dff33b-48f5-42fd-962e-942d8b0a6678"	SingleSelect
16	"206cb04a-9519-4623-8109-8b6cf43f224e"	SingleSelect
17	"db098389-9e70-4b6b-ab75-8155ac596158"	SingleSelect
18	"145591d2-2d08-42da-b48e-60c615c93ef0"	SingleSelect
19	"ba7ff179-0a65-4ad2-849f-06b02425355c"	SingleSelect
20	"2cb637a8-25d5-4711-b237-d0f30a4fc69e"	SingleSelect
21	"10e312b7-1324-4335-8e82-ba0db197f99f"	MutliSelect
22	"e0c8ecf3-ea97-455a-932a-0b4706f8a620"	SingleSelect
23	"6cd4d59a-c733-423f-8348-d59f652eb6e5"	SingleSelect
24	"9c71a15c-79ff-4981-adbf-41a116e59126"	MutliSelect
25	"7f3e04bb-5e0d-48f6-a9ac-2f210aec9f97"	MutliSelect
26	"1450e285-a0bf-4306-bcb0-964b38720b9c"	MutliSelect
27	"8723124d-771b-4717-8e32-d49189a81473"	MutliSelect
28	"0dfd73bd-6c6e-400c-a036-0ec64085cbeb"	OpenText

	SubjectName \
0	Did you buy Chickapea pasta after trying the s...
1	Have you ever tried legume-based pasta before?...
2	How many boxes of Chickapea pasta did you buy?
3	User Distribution
4	Province
5	Gender
6	Do you have any kids?
7	Country
8	How old are your kids?
9	Age Range

10 Which of the following stores can you easily a...
 11 How do you rate each of these characteristics ...
 12 How do you rate each of these characteristics ...
 13 How do you rate each of these characteristics ...
 14 How do you rate each of these characteristics ...
 15 How do you rate each of these characteristics ...
 16 How do you rate the taste of the pasta?
 17 How do you rate the texture of the pasta?
 18 How does the taste and texture of Chickapea pa...
 19 How often are you planning to purchase Chickap...
 20 How often do you eat pasta?
 21 If not planning to buy it at least once a mont...
 22 On a scale of 0 to 10, how likely are you to r...
 23 Purchase Intent
 24 What could be improved about Chickapea Spirals?
 25 What did you like the most about Chickapea Spi...
 26 Where did you buy Chickapea pasta?
 27 Which brand(s) comes to mind when you think ab...
 28 Please provide a brief review of the pasta you...

	SubjectValues
0	["No","Yes"]
1	["No","Yes"]
2	["1","2-3","4-5","More than 5"]
3	["Activation (Dormant)","Acquisition (New-to-b...]
4	["Alabama","US","Alaska","US","Alberta","CA","...]
5	["Female","I would rather not say","Male","Non...]
6	["Expecting Children","No","Yes"]
7	["CA","US"]
8	["0 to 1 years old","1 to 3 years old","3 to 5...]
9	["20-30","31-40","41-50","51-60","61-70","Unde...]
10	["Albertsons","H-E-B","Kroger","Meijer","Other...]
11	["😊","😊","😊","😊"]
12	["😊","😊","😊","😊"]
13	["😊","😊","😊","😊"]
14	["😊","😊","😊","😊"]
15	["😊","😊","😊","😊"]
16	["1","2","3","4","5"]
17	["1","2","3","4","5"]
18	["About the same","Better","This was my first ...]
19	["I'm not going to buy it","Once a month","Onc...]
20	["A couple of times a month","More than 2 time...]
21	["Didn't like the taste or texture","I eat pas...]
22	["0","1","10","2","3","4","5","6","7","8","9"]
23	["No","Yes"]
24	["Flavor (Taste)","Ingredients","Nothing (it w...]
25	["Flavor (Taste)","Ingredients","Nothing","Nut...]
26	["Amazon","Brands's website","Fresh Thyme","Ha...]

```
27 ["Banza","Barilla Protein+","Brami","Chickapea...
28                                     NaN
```

```
choice
```

```
                                RecordId \
0      "ac53fafb-7db0-4a5b-a8d6-2c18fab493fb"
1      "57e7cbef-0cc4-45cf-9cb3-644ec73b59db"
2      "24e414a4-9972-4314-b513-26a52df4eb3d"
3      "fa99ef64-dcb9-4893-9df4-ce1dadab259da"
4      "a8daeeaa2-d746-4cfb-a3df-570451a2a833"
...
24250  "d8712e44-6d34-4542-bf7d-ae85ae9936ae"
24251  "7ecb178d-6043-41dd-b758-e2978098a388"
24252  "39b4d277-dc1f-4684-babb-7dc4ac823e13"
24253  "0faeb424-c84e-4ece-9645-97960c6e861c"
24254  "662a0e45-6d00-4f4f-a880-ae459c2c693f"
```

```
                                SamplerId \
0      "0f98a10f-f296-4808-a7ec-2d20c60bb1cf"
1      "ae3f98ed-cee7-4645-80f2-bb4c620e9897"
2      "92e80710-705b-48b3-ac62-962005cc254b"
3      "d03f6cf3-5a10-4d9f-8922-6033f3a39688"
4      "4d8be7ea-65a4-427f-8e77-368883fe5193"
...
24250  "b310ed8c-9610-4079-8b43-d70d678ac7ba"
24251  "3c5f356a-fc6f-4873-aadf-6099eb3d10cb"
24252  "ff11c90d-3152-4d2c-85cc-0da30b6b119c"
24253  "bae1a189-3e4a-489c-b462-c4b20400d0e4"
24254  "ddb13109-9906-411b-b345-2602d4a255fa"
```

```
                                CampaignSubjectId SubjectValue
0      "1450e285-a0bf-4306-bcb0-964b38720b9c"      Kroger
1      "1450e285-a0bf-4306-bcb0-964b38720b9c"      Other
2      "206cb04a-9519-4623-8109-8b6cf43f224e"          4
3      "7f3e04bb-5e0d-48f6-a9ac-2f210aec9f97"  Ingredients
4      "7f3e04bb-5e0d-48f6-a9ac-2f210aec9f97"      Texture
...
24250  "347c0747-787f-4d39-8176-e076649d80fa"      Yes
24251  "347c0747-787f-4d39-8176-e076649d80fa"      Yes
24252  "347c0747-787f-4d39-8176-e076649d80fa"      Yes
24253  "347c0747-787f-4d39-8176-e076649d80fa"      Yes
24254  "347c0747-787f-4d39-8176-e076649d80fa"      Yes
```

```
[24255 rows x 4 columns]
```

```
# Now we want to extract the data from choice by using the campaign datas
```

CampaignSubjectId datas in choice df is similiar with Id in campaign df

```
campaign = campaign.rename(columns={'Id': 'CampaignSubjectId'})
campaign
```

	CampaignSubjectId	SubjectType \
0	"f36cb6ff-8df4-4842-b59f-d05153bbb560"	SingleSelect
1	"d7f7cebb-baa8-41ce-95c5-688f86e50a75"	SingleSelect
2	"80011846-c284-4aa7-9742-8c3f16e131b7"	SingleSelect
3	"3c327491-9c15-4ab3-bdb9-8d2aafbbf87e"	SingleSelect
4	"e0b39b39-682d-462d-ab23-8de2577628fe"	LocationData
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6	"ecb57a0f-3779-4ded-99a7-1fa4d9c6a14f"	SingleSelect
7	"90c36fb3-1cf8-4659-b2bd-6d1dce401ba0"	LocationData
8	"8e630a29-7f74-41f8-9023-813459d7e254"	SingleSelect
9	"90f8fee2-9939-4dc3-a91c-2da82a2f8053"	SingleSelect
10	"e4eeb58a-5c80-428d-bf24-7372867dd28d"	MutliSelect
11	"935791bc-4831-4386-aa99-90f3c3ddb571"	SingleSelect
12	"044a1964-ec46-4404-9089-120f41dd0f54"	SingleSelect
13	"e8e6cdbe-9500-4eef-a117-1a31c5c35990"	SingleSelect
14	"8ca14928-2fbe-42b6-8713-5b4e17459f80"	SingleSelect
15	"a3dff33b-48f5-42fd-962e-942d8b0a6678"	SingleSelect
16	"206cb04a-9519-4623-8109-8b6cf43f224e"	SingleSelect
17	"db098389-9e70-4b6b-ab75-8155ac596158"	SingleSelect
18	"145591d2-2d08-42da-b48e-60c615c93ef0"	SingleSelect
19	"ba7ff179-0a65-4ad2-849f-06b02425355c"	SingleSelect
20	"2cb637a8-25d5-4711-b237-d0f30a4fc69e"	SingleSelect
21	"10e312b7-1324-4335-8e82-ba0db197f99f"	MutliSelect
22	"e0c8ecf3-ea97-455a-932a-0b4706f8a620"	SingleSelect
23	"6cd4d59a-c733-423f-8348-d59f652eb6e5"	SingleSelect
24	"9c71a15c-79ff-4981-adbf-41a116e59126"	MutliSelect
25	"7f3e04bb-5e0d-48f6-a9ac-2f210aec9f97"	MutliSelect
26	"1450e285-a0bf-4306-bcb0-964b38720b9c"	MutliSelect
27	"8723124d-771b-4717-8e32-d49189a81473"	MutliSelect
28	"0dfd73bd-6c6e-400c-a036-0ec64085cbeb"	OpenText

	SubjectName \
0	Did you buy Chickapea pasta after trying the s...
1	Have you ever tried legume-based pasta before?...
2	How many boxes of Chickapea pasta did you buy?
3	User Distribution
4	Province
5	Gender
6	Do you have any kids?
7	Country
8	How old are your kids?
9	Age Range
10	Which of the following stores can you easily a...
11	How do you rate each of these characteristics ...
12	How do you rate each of these characteristics ...

13 How do you rate each of these characteristics ...
 14 How do you rate each of these characteristics ...
 15 How do you rate each of these characteristics ...
 16 How do you rate the taste of the pasta?
 17 How do you rate the texture of the pasta?
 18 How does the taste and texture of Chickapea pa...
 19 How often are you planning to purchase Chickap...
 20 How often do you eat pasta?
 21 If not planning to buy it at least once a mont...
 22 On a scale of 0 to 10, how likely are you to r...
 23 Purchase Intent
 24 What could be improved about Chickapea Spirals?
 25 What did you like the most about Chickapea Spi...
 26 Where did you buy Chickapea pasta?
 27 Which brand(s) comes to mind when you think ab...
 28 Please provide a brief review of the pasta you...

	SubjectValues
0	["No","Yes"]
1	["No","Yes"]
2	["1","2-3","4-5","More than 5"]
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6	["Expecting Children","No","Yes"]
7	["CA","US"]
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9	["20-30","31-40","41-50","51-60","61-70","Unde...]
10	["Albertsons","H-E-B","Kroger","Meijer","Other...]
11	["😊","😊","😊","😊"]
12	["😊","😊","😊","😊"]
13	["😊","😊","😊","😊"]
14	["😊","😊","😊","😊"]
15	["😊","😊","😊","😊"]
16	["1","2","3","4","5"]
17	["1","2","3","4","5"]
18	["About the same","Better","This was my first ...]
19	["I'm not going to buy it","Once a month","Onc...]
20	["A couple of times a month","More than 2 time...]
21	["Didn't like the taste or texture","I eat pas...]
22	["0","1","10","2","3","4","5","6","7","8","9"]
23	["No","Yes"]
24	["Flavor (Taste)","Ingredients","Nothing (it w...]
25	["Flavor (Taste)","Ingredients","Nothing","Nut...]
26	["Amazon","Brands's website","Fresh Thyme","Ha...]
27	["Banza","Barilla Protein+","Brami","Chickapea...]
28	NaN

It is Id of questions (Features) in choice df and its SubjectValue are the related answers.

Now we want to replace the features in 'CampaignSubjectId' in choice df

```
choice = choice.merge(
    campaign[['CampaignSubjectId', 'SubjectName']],
    on= 'CampaignSubjectId',
    how= 'left'
)
choice
```

	RecordId \	
0	"ac53fafb-7db0-4a5b-a8d6-2c18fab493fb"	
1	"57e7cbef-0cc4-45cf-9cb3-644ec73b59db"	
2	"24e414a4-9972-4314-b513-26a52df4eb3d"	
3	"fa99ef64-dcb9-4893-9df4-ce1dadab259da"	
4	"a8daeeaa2-d746-4cfb-a3df-570451a2a833"	
...	...	
24250	"d8712e44-6d34-4542-bf7d-ae85ae9936ae"	
24251	"7ecb178d-6043-41dd-b758-e2978098a388"	
24252	"39b4d277-dc1f-4684-babb-7dc4ac823e13"	
24253	"0faeb424-c84e-4ece-9645-97960c6e861c"	
24254	"662a0e45-6d00-4f4f-a880-ae459c2c693f"	

	SamplerId \	
0	"0f98a10f-f296-4808-a7ec-2d20c60bb1cf"	
1	"ae3f98ed-cee7-4645-80f2-bb4c620e9897"	
2	"92e80710-705b-48b3-ac62-962005cc254b"	
3	"d03f6cf3-5a10-4d9f-8922-6033f3a39688"	
4	"4d8be7ea-65a4-427f-8e77-368883fe5193"	
...	...	
24250	"b310ed8c-9610-4079-8b43-d70d678ac7ba"	
24251	"3c5f356a-fc6f-4873-aadf-6099eb3d10cb"	
24252	"ff11c90d-3152-4d2c-85cc-0da30b6b119c"	
24253	"bae1a189-3e4a-489c-b462-c4b20400d0e4"	
24254	"ddb13109-9906-411b-b345-2602d4a255fa"	

	CampaignSubjectId	SubjectValue \
0	"1450e285-a0bf-4306-bcb0-964b38720b9c"	Kroger
1	"1450e285-a0bf-4306-bcb0-964b38720b9c"	Other
2	"206cb04a-9519-4623-8109-8b6cf43f224e"	4
3	"7f3e04bb-5e0d-48f6-a9ac-2f210aec9f97"	Ingredients
4	"7f3e04bb-5e0d-48f6-a9ac-2f210aec9f97"	Texture
...	...	...
24250	"347c0747-787f-4d39-8176-e076649d80fa"	Yes
24251	"347c0747-787f-4d39-8176-e076649d80fa"	Yes
24252	"347c0747-787f-4d39-8176-e076649d80fa"	Yes

24253	"347c0747-787f-4d39-8176-e076649d80fa"	Yes
24254	"347c0747-787f-4d39-8176-e076649d80fa"	Yes

		SubjectName
0	Where did you buy Chickapea pasta?	
1	Where did you buy Chickapea pasta?	
2	How do you rate the taste of the pasta?	
3	What did you like the most about Chickapea Spi...	
4	What did you like the most about Chickapea Spi...	
...		...
24250		NaN
24251		NaN
24252		NaN
24253		NaN
24254		NaN

[24255 rows x 5 columns]

In above we have some 'CampaignSubjectId' datas which their 'SubjectName' is Nan. It means the features are not defined. So we need to remove them.

```
choice = choice[choice['SubjectName'].notna()]
choice
```

	RecordId \
0	"ac53fafb-7db0-4a5b-a8d6-2c18fab493fb"
1	"57e7cbef-0cc4-45cf-9cb3-644ec73b59db"
2	"24e414a4-9972-4314-b513-26a52df4eb3d"
3	"fa99ef64-dcb9-4893-9df4-ce1dadab259da"
4	"a8daeeaa2-d746-4cfb-a3df-570451a2a833"
...	...
23230	"841f48b7-0163-4985-8d28-9ae92311a0ee"
23231	"9f2caa82-5d28-47c5-9acd-99429cbb4be3"
23232	"640cd1e8-630b-40a5-863c-cfcc3dfdbd27"
23233	"1cc6c2c5-a024-4113-a95b-326cca7bc1ca"
23234	"0bb58e36-663b-4cbc-9861-ab1053eacc11"

	SamplerId \
0	"0f98a10f-f296-4808-a7ec-2d20c60bb1cf"
1	"ae3f98ed-cee7-4645-80f2-bb4c620e9897"
2	"92e80710-705b-48b3-ac62-962005cc254b"
3	"d03f6cf3-5a10-4d9f-8922-6033f3a39688"
4	"4d8be7ea-65a4-427f-8e77-368883fe5193"
...	...
23230	"8037afdb-c0ae-4d4f-b667-8616e80188b7"
23231	"13c887ba-fef7-4622-a343-7c2259d26736"
23232	"8f798105-ee3f-4a53-928b-c1e04d6de132"
23233	"cad687f8-b109-4284-b3a4-f2349413e12a"

```

23234 "2ac24228-101d-4ec5-90e2-1b95c56f019c"
      CampaignSubjectId \
0      "1450e285-a0bf-4306-bcb0-964b38720b9c"
1      "1450e285-a0bf-4306-bcb0-964b38720b9c"
2      "206cb04a-9519-4623-8109-8b6cf43f224e"
3      "7f3e04bb-5e0d-48f6-a9ac-2f210aec9f97"
4      "7f3e04bb-5e0d-48f6-a9ac-2f210aec9f97"
...
23230 "145591d2-2d08-42da-b48e-60c615c93ef0"
23231 "145591d2-2d08-42da-b48e-60c615c93ef0"
23232 "145591d2-2d08-42da-b48e-60c615c93ef0"
23233 "145591d2-2d08-42da-b48e-60c615c93ef0"
23234 "145591d2-2d08-42da-b48e-60c615c93ef0"
      SubjectValue \
0      Kroger
1      Other
2      4
3      Ingredients
4      Texture
...
23230 This was my first chickpea pasta
23231 This was my first chickpea pasta
23232 This was my first chickpea pasta
23233 This was my first chickpea pasta
23234 This was my first chickpea pasta
      SubjectName
0      Where did you buy Chickapea pasta?
1      Where did you buy Chickapea pasta?
2      How do you rate the taste of the pasta?
3      What did you like the most about Chickapea Spi...
4      What did you like the most about Chickapea Spi...
...
23230 How does the taste and texture of Chickapea pa...
23231 How does the taste and texture of Chickapea pa...
23232 How does the taste and texture of Chickapea pa...
23233 How does the taste and texture of Chickapea pa...
23234 How does the taste and texture of Chickapea pa...
[23234 rows x 5 columns]

```

Now I want to separate the SamplerId and put SubjectName as feature with related SubjectValue.

```

main_df = (
    choice
    .pivot_table(

```



```

        index="SamplerId",
        columns="SubjectName",
        values="SubjectValue",
        aggfunc="first"
    )
    .reset_index()
)
main_df

```

SubjectName \	SamplerId	Age Range	Country
0	"00252e52-9e6f-4dc2-821e-0a2d0d0b5940"	41-50	US
1	"0067fb06-29d8-4bc4-93b2-4cf1e505d122"	51-60	US
2	"009c1337-4a4e-44b0-a130-a824cdc0dc6d"	41-50	US
3	"012cae1c-db14-469a-862e-d72e21ffdee1"	41-50	US
4	"01860c38-9987-48d9-871c-69a1272f49a1"	51-60	US
..	...	...	...
726	"fd6eb01a-3f2d-4e6d-85db-c11ec2d99d40"	41-50	US
727	"fd94de0d-a80f-486e-9d3e-1aeb6fc289b9"	51-60	US
728	"feb1699c-ad00-49f4-881b-9c101fdc06c5"	51-60	US
729	"feb2c322-5627-43c7-8810-ccb37c07d76c"	20-30	US
730	"ff3db08e-b008-43ad-87b5-64b46fbf75c3"	31-40	US

SubjectName	Did you buy Chickapea pasta after trying the sample?	\
0	No	
1	No	
2	No	
3	No	
4	Yes	
..	...	
726	No	
727	No	
728	No	
729	No	
730	Yes	

SubjectName	Do you have any kids?	Gender	\
0	No	Female	
1	Yes	Female	
2	Yes	Female	

3	Yes	Female
4	Yes	Female
..	...	...
726	Yes	Male
727	No	Female
728	Yes	Female
729	Yes	Female
730	Yes	Female

SubjectName Have you ever tried legume-based pasta before? (chickpea, lentil, pea, bean, etc.) \

0	Yes
1	Yes
2	No
3	Yes
4	No
..	...
726	Yes
727	Yes
728	Yes
729	No
730	Yes

SubjectName How do you rate each of these characteristics of Chickapea packaging?: Clarity of information \

0	😊
1	😊
2	😊
3	😊
4	😊
..	...
726	😊
727	😊

728	😊
-----	---

729	😊
-----	---

730	😊
-----	---

SubjectName How do you rate each of these characteristics of Chickapea packaging?: Ease of use (Open, reseal, and handle) \

0	😊
---	---

1	😊
---	---

2	😊
---	---

3	😊
---	---

4	😊
---	---

..	...
----	-----

726	😊
-----	---

727	😊
-----	---

728	😊
-----	---

729	😊
-----	---

730	😊
-----	---

SubjectName How do you rate each of these characteristics of Chickapea packaging?: Overall packaging \

0	😊
---	---

1	😊
---	---

2	😊
---	---

3	😊
---	---

4	😊
---	---

..	...
----	-----

726	😊
-----	---

727	😊
-----	---

728	😊
-----	---

729	😊
-----	---

730	😐
-----	---

SubjectName	...	If not planning to buy it at least once a month, why not? \
-------------	-----	---

0	...	NaN
---	-----	-----

1	...	NaN
---	-----	-----

2	...	NaN
---	-----	-----

3	...	NaN
---	-----	-----

4	...	NaN
---	-----	-----

..	...	...
----	-----	-----

726	...	NaN
-----	-----	-----

727	...	NaN
-----	-----	-----

728	...	NaN
-----	-----	-----

729	...	NaN
-----	-----	-----

730	...	NaN
-----	-----	-----

SubjectName	On a scale of 0 to 10, how likely are you to recommend Chickapea pasta to a friend or colleague? \
-------------	--

0	9
---	---

1	10
---	----

2	7
---	---

3	8
---	---

4	10
---	----

..	...
----	-----

726	6
-----	---

727	10
-----	----

728	9
-----	---

729	10
-----	----

730

6

SubjectName	Province	Purchase Intent	User
Distribution \			
0	California, US	Yes	Acquisition (New-to-brand)
1	Illinois, US	Yes	Activation
(Dormant)			
2	California, US	Yes	Acquisition (New-to-brand)
3	Arizona, US	Yes	Acquisition (New-to-brand)
4	California, US	Yes	Acquisition (New-to-brand)
..	...	...	
...			
726	Georgia, US	Yes	Acquisition (New-to-brand)
727	Ohio, US	Yes	Acquisition (New-to-brand)
728	California, US	Yes	Activation
(Dormant)			
729	South Carolina, US	Yes	Acquisition (New-to-brand)
730	Florida, US	Yes	Acquisition (New-to-brand)

SubjectName	What could be improved about Chickapea Spirals?	
0	Price (\$3.99/box)	\
1	Price (\$3.99/box)	
2	Flavor (Taste)	
3	Texture	
4	NaN	
..	...	
726	Texture	
727	Price (\$3.99/box)	
728	Price (\$3.99/box)	
729	NaN	
730	Price (\$3.99/box)	

SubjectName	What did you like the most about Chickapea Spirals?	
0	Ingredients	\
1	Flavor (Taste)	
2	Price (\$3.99/box)	
3	Flavor (Taste)	
4	Ingredients	
..	...	
726	Nutritional benefits	

727		Texture
728	Nutritional	benefits
729		Texture
730		Texture

SubjectName Where did you buy Chickapea pasta? \

0	NaN
1	NaN
2	NaN
3	NaN
4	Other
..	...
726	Amazon
727	Fresh Thyme
728	NaN
729	NaN
730	Publix

SubjectName Which brand(s) comes to mind when you think about high protein pasta? \

0	Veggiecraft Farms
1	Brami
2	Chickapea
3	Barilla Protein+
4	None of the above
..	...
726	Barilla Protein+
727	Other
728	Chickapea
729	Veggiecraft Farms
730	Chickapea

SubjectName Which of the following stores can you easily access?

0	Whole Foods Market
1	Whole Foods Market
2	Albertsons
3	Albertsons
4	Sprouts
..	...
726	Other

727	Meijer
728	Walmart
729	Walmart
730	Publix

[731 rows x 29 columns]

Finally Our main_df is created. Lets save it.

```
main_df.to_csv('Main DF.csv', index=False)
```

As some of the features are long, it should be better to replace the small and suitable words to them

```
rename_features = {
    "Did you buy Chickapea pasta after trying the sample?": "Bought After sample",
    "Do you have any kids?": "Has Kids",
    "How old are your kids?": "Kids Year",
    "Have you ever tried legume-based pasta before? (chickpea, lentil, pea, bean, etc.)": "Tried Legume Pasta",
    "On a scale of 0 to 10, how likely are you to recommend Chickapea pasta to a friend or colleague?": "Pasta Recommendation Scale",

    "How do you rate each of these characteristics of Chickapea packaging?: Clarity of information": "Clarity Packaging",
    "How do you rate each of these characteristics of Chickapea packaging?: Ease of use (Open, reseal, and handle)": "Ease Packaging",
    "How do you rate each of these characteristics of Chickapea packaging?: Overall packaging": "Overall Packaging",
    "How do you rate each of these characteristics of Chickapea packaging?: Sustainability": "Sustainability Packaging",
    "How do you rate each of these characteristics of Chickapea packaging?: Visual appeal": "Visual Packaging",

    "If not planning to buy it at least once a month, why not?": "Not Monthly Buy",
    "What could be improved about Chickapea Spirals?": "Improve Areas",
    "What did you like the most about Chickapea Spirals?": "Liked Most",
    "How many boxes of Chickapea pasta did you buy?": "Pasta Boxes",

    "Where did you buy Chickapea pasta?": "Pasta Channel",
    "Which brand(s) comes to mind when you think about high protein pasta?": "Brand Recall",
    "Which of the following stores can you easily access?": "Easy Access Store",
}
```

```

    "How do you rate the taste of the pasta?": "Taste Rate",
    "How do you rate the texture of the pasta?": "Texture Rate",
    "How does the taste and texture of Chickapea pasta compare to
other chickpea-based pasta you tried?" : "Taste Texture Comparision",
    "How often are you planning to purchase Chickapea pasta in the
future?" : "Purchase Planning",
    "How often do you eat pasta?": "Pasta Usage",
    "Please provide a brief review of the pasta you had from
Chickapea.": "Review"
}

```

```

main_df = main_df.rename(columns=rename_features)
main_df

```

SubjectName	SamplerId	Age Range	Country
0	"00252e52-9e6f-4dc2-821e-0a2d0d0b5940"	41-50	US
1	"0067fb06-29d8-4bc4-93b2-4cf1e505d122"	51-60	US
2	"009c1337-4a4e-44b0-a130-a824cdc0dc6d"	41-50	US
3	"012cae1c-db14-469a-862e-d72e21ffdee1"	41-50	US
4	"01860c38-9987-48d9-871c-69a1272f49a1"	51-60	US
..	...	...	...
726	"fd6eb01a-3f2d-4e6d-85db-c11ec2d99d40"	41-50	US
727	"fd94de0d-a80f-486e-9d3e-1aeb6fc289b9"	51-60	US
728	"feb1699c-ad00-49f4-881b-9c101fdc06c5"	51-60	US
729	"feb2c322-5627-43c7-8810-ccb37c07d76c"	20-30	US
730	"ff3db08e-b008-43ad-87b5-64b46fbf75c3"	31-40	US

SubjectName	Bought After sample	Has Kids	Gender	Tried Legume Pasta	\
0	No	No	Female	Yes	
1	No	Yes	Female	Yes	
2	No	Yes	Female	No	
3	No	Yes	Female	Yes	
4	Yes	Yes	Female	No	
..	...	...	...	...	
726	No	Yes	Male	Yes	
727	No	No	Female	Yes	
728	No	Yes	Female	Yes	
729	No	Yes	Female	No	
730	Yes	Yes	Female	Yes	

SubjectName	Clarity	Packaging	Ease	Packaging	Overall	Packaging	...	\
0		😊		😊		😊	...	
1		😊		😊		😊	...	
2		😊		😊		😊	...	
3		😊		😊		😊	...	
4		😊		😊		😊	...	
..		...		...		...	...	
726		😊		😊		😊	...	
727		😊		😊		😊	...	
728		😊		😊		😊	...	
729		😊		😊		😊	...	
730		😊		😊		😊	...	

SubjectName	Not	Monthly	Buy	Pasta	Recommendation	Scale
Province \						
0			NaN			9
California, US						
1			NaN			10
Illinois, US						
2			NaN			7
California, US						
3			NaN			8
Arizona, US						
4			NaN			10
California, US						
..			...			...
...						
726			NaN			6
Georgia, US						
727			NaN			10
Ohio, US						
728			NaN			9
California, US						
729			NaN			10
Carolina, US						South
730			NaN			6
Florida, US						

SubjectName	Purchase	Intent	User	Distribution	Improve
Areas \					

0		Yes	Acquisition (New-to-brand)	Price
1	(\$3.99/box)	Yes	Activation (Dormant)	Price
2		Yes	Acquisition (New-to-brand)	Flavor
3	(Taste)	Yes	Acquisition (New-to-brand)	
4	Texture	Yes	Acquisition (New-to-brand)	
NaN				
..		...		...
...				
726		Yes	Acquisition (New-to-brand)	
727	Texture	Yes	Acquisition (New-to-brand)	Price
728	(\$3.99/box)	Yes	Activation (Dormant)	Price
729	(\$3.99/box)	Yes	Acquisition (New-to-brand)	
NaN				
730		Yes	Acquisition (New-to-brand)	Price
	(\$3.99/box)			

SubjectName	Liked Most	Pasta	Channel	Brand Recall	\
0	Ingredients		NaN	Veggiecraft Farms	
1	Flavor (Taste)		NaN	Brami	
2	Price (\$3.99/box)		NaN	Chickapea	
3	Flavor (Taste)		NaN	Barilla Protein+	
4	Ingredients		Other	None of the above	
..		...	...	...	
726	Nutritional benefits		Amazon	Barilla Protein+	
727	Texture	Fresh	Thyme	Other	
728	Nutritional benefits		NaN	Chickapea	
729	Texture		NaN	Veggiecraft Farms	
730	Texture		Publix	Chickapea	

SubjectName	Easy Access Store
0	Whole Foods Market
1	Whole Foods Market
2	Albertsons
3	Albertsons
4	Sprouts
..	...
726	Other
727	Meijer
728	Walmart
729	Walmart
730	Publix

[731 rows x 29 columns]

As I checked, there are some NaNs. We can rearrange or remove the NaNs, if needed.

```
main_df.isna().sum()

SubjectName
SamplerId           0
Age Range           0
Country             0
Bought After sample 0
Has Kids            0
Gender              0
Tried Legume Pasta  0
Clarity Packaging    3
Ease Packaging       3
Overall Packaging    0
Sustainability Packaging 3
Visual Packaging     0
Taste Rate          0
Texture Rate        0
Taste Texture Comparision 0
Pasta Boxes         406
Purchase Planning   0
Pasta Usage         0
Kids Year           0
Not Monthly Buy     664
Pasta Recommendation Scale 0
Province            0
Purchase Intent     0
User Distribution   0
Improve Areas       6
Liked Most          0
Pasta Channel       406
Brand Recall        0
Easy Access Store   0
dtype: int64
```

Also we will change the emojis and 'Yes/No' to numerical datas.

```
emoji_cols = [
    "Clarity Packaging",
    "Ease Packaging",
    "Overall Packaging",
    "Sustainability Packaging",
    "Visual Packaging"
]

emoji_map = {
    "😊": 4,
    "😊": 3,
```

```

    "😄": 2,
    "😞": 1
}

for col in emoji_cols:
    main_df[col] = main_df[col].map(emoji_map).astype("Int64")

```

main_df

SubjectName \	SamplerId	Age Range	Country
0	"00252e52-9e6f-4dc2-821e-0a2d0d0b5940"	41-50	US
1	"0067fb06-29d8-4bc4-93b2-4cf1e505d122"	51-60	US
2	"009c1337-4a4e-44b0-a130-a824cdc0dc6d"	41-50	US
3	"012cae1c-db14-469a-862e-d72e21ffdee1"	41-50	US
4	"01860c38-9987-48d9-871c-69a1272f49a1"	51-60	US
..	...	...	...
726	"fd6eb01a-3f2d-4e6d-85db-c11ec2d99d40"	41-50	US
727	"fd94de0d-a80f-486e-9d3e-1aeb6fc289b9"	51-60	US
728	"feb1699c-ad00-49f4-881b-9c101fdc06c5"	51-60	US
729	"feb2c322-5627-43c7-8810-ccb37c07d76c"	20-30	US
730	"ff3db08e-b008-43ad-87b5-64b46fbf75c3"	31-40	US

SubjectName	Bought After sample	Has Kids	Gender	Tried Legume	Pasta \
0	No	No	Female		Yes
1	No	Yes	Female		Yes
2	No	Yes	Female		No
3	No	Yes	Female		Yes
4	Yes	Yes	Female		No
..	...	...	...		...
726	No	Yes	Male		Yes
727	No	No	Female		Yes
728	No	Yes	Female		Yes
729	No	Yes	Female		No
730	Yes	Yes	Female		Yes

SubjectName	Clarity Packaging	Ease Packaging	Overall Packaging	...
\				
0	4	4	3	...

1	4	4	4	...
2	4	4	4	...
3	4	4	3	...
4	4	4	4	...
..	...	...	...	...
726	3	3	3	...
727	4	4	4	...
728	4	4	3	...
729	4	4	4	...
730	3	3	1	...

SubjectName	Not Monthly Buy	Pasta Recommendation	Scale
Province \			
0	NaN		9
California, US			
1	NaN		10
Illinois, US			
2	NaN		7
California, US			
3	NaN		8
Arizona, US			
4	NaN		10
California, US			
..	...		...
...			
726	NaN		6
Georgia, US			
727	NaN		10
Ohio, US			
728	NaN		9
California, US			
729	NaN		10
South Carolina, US			
730	NaN		6
Florida, US			

SubjectName	Purchase Intent	User Distribution	Improve
Areas \			
0	Yes	Acquisition (New-to-brand)	Price
(\$3.99/box)			

1		Yes	Activation (Dormant)	Price
2	(\$3.99/box)	Yes	Acquisition (New-to-brand)	Flavor
3	(Taste)	Yes	Acquisition (New-to-brand)	
4	Texture	Yes	Acquisition (New-to-brand)	
NaN				
..		...		...
...				
726		Yes	Acquisition (New-to-brand)	
727	Texture	Yes	Acquisition (New-to-brand)	Price
728	(\$3.99/box)	Yes	Activation (Dormant)	Price
729	(\$3.99/box)	Yes	Acquisition (New-to-brand)	
NaN				
730		Yes	Acquisition (New-to-brand)	Price
	(\$3.99/box)			

SubjectName	Liked Most Pasta Channel	Brand Recall \
0	Ingredients	NaN Veggiecraft Farms
1	Flavor (Taste)	NaN Brami
2	Price (\$3.99/box)	NaN Chickapea
3	Flavor (Taste)	NaN Barilla Protein+
4	Ingredients	Other None of the above
..	...	...
726	Nutritional benefits	Amazon Barilla Protein+
727	Texture	Fresh Thyme Other
728	Nutritional benefits	NaN Chickapea
729	Texture	NaN Veggiecraft Farms
730	Texture	Publix Chickapea

SubjectName	Easy Access Store
0	Whole Foods Market
1	Whole Foods Market
2	Albertsons
3	Albertsons
4	Sprouts
..	...
726	Other
727	Meijer
728	Walmart
729	Walmart
730	Publix

[731 rows x 29 columns]

```

boolean_cols = [
    "Bought After sample",
    "Has Kids",
    "Tried Legume Pasta",
    "Purchase Intent"
]

boolean_map = {"No": 0, "Yes": 1}

for col in boolean_cols:
    main_df[col] = main_df[col].map(boolean_map).astype("Int64")

```

main_df

SubjectName \	SamplerId	Age Range	Country
0	"00252e52-9e6f-4dc2-821e-0a2d0d0b5940"	41-50	US
1	"0067fb06-29d8-4bc4-93b2-4cf1e505d122"	51-60	US
2	"009c1337-4a4e-44b0-a130-a824cdc0dc6d"	41-50	US
3	"012cae1c-db14-469a-862e-d72e21ffdee1"	41-50	US
4	"01860c38-9987-48d9-871c-69a1272f49a1"	51-60	US
..	...	...	...
726	"fd6eb01a-3f2d-4e6d-85db-c11ec2d99d40"	41-50	US
727	"fd94de0d-a80f-486e-9d3e-1aeb6fc289b9"	51-60	US
728	"feb1699c-ad00-49f4-881b-9c101fdc06c5"	51-60	US
729	"feb2c322-5627-43c7-8810-ccb37c07d76c"	20-30	US
730	"ff3db08e-b008-43ad-87b5-64b46fbf75c3"	31-40	US

SubjectName \	Bought After sample	Has Kids	Gender	Tried Legume Pasta
0	0	0	Female	1
1	0	1	Female	1
2	0	1	Female	0
3	0	1	Female	1
4	1	1	Female	0

..	...	...	...	...
726	0	1	Male	1
727	0	0	Female	1
728	0	1	Female	1
729	0	1	Female	0
730	1	1	Female	1
SubjectName Clarity Packaging Ease Packaging Overall Packaging ...				
\				
0	4	4	3	...
1	4	4	4	...
2	4	4	4	...
3	4	4	3	...
4	4	4	4	...
..	...	...	...	...
726	3	3	3	...
727	4	4	4	...
728	4	4	3	...
729	4	4	4	...
730	3	3	1	...
SubjectName Not Monthly Buy Pasta Recommendation Scale				
Province \				
0	NaN		9	
California, US				
1	NaN		10	
Illinois, US				
2	NaN		7	
California, US				
3	NaN		8	
Arizona, US				
4	NaN		10	
California, US				
..	...		...	

...		
726	NaN	6
Georgia, US		
727	NaN	10
Ohio, US		
728	NaN	9
California, US		
729	NaN	10
Carolina, US		
730	NaN	6
Florida, US		

SubjectName	Purchase Intent	User Distribution	Improve Areas \
0	1	Acquisition (New-to-brand)	Price (\$3.99/box)
1	1	Activation (Dormant)	Price (\$3.99/box)
2	1	Acquisition (New-to-brand)	Flavor (Taste)
3	1	Acquisition (New-to-brand)	Texture
4	1	Acquisition (New-to-brand)	NaN

..	...	...
...		
726	1	Acquisition (New-to-brand)
Texture		
727	1	Acquisition (New-to-brand)
Price (\$3.99/box)		
728	1	Activation (Dormant)
Price (\$3.99/box)		
729	1	Acquisition (New-to-brand)
NaN		
730	1	Acquisition (New-to-brand)
Price (\$3.99/box)		

SubjectName	Liked Most	Pasta Channel	Brand Recall \
0	Ingredients	NaN	Veggiecraft Farms
1	Flavor (Taste)	NaN	Brami
2	Price (\$3.99/box)	NaN	Chickapea
3	Flavor (Taste)	NaN	Barilla Protein+
4	Ingredients	Other	None of the above
..	...	...	...
726	Nutritional benefits	Amazon	Barilla Protein+
727	Texture	Fresh Thyme	Other
728	Nutritional benefits	NaN	Chickapea
729	Texture	NaN	Veggiecraft Farms
730	Texture	Publix	Chickapea

```

SubjectName  Easy Access Store
0           Whole Foods Market
1           Whole Foods Market
2                Albertsons
3                Albertsons
4                Sprouts
..          ...
726          Other
727          Meijer
728          Walmart
729          Walmart
730          Publix

```

```
[731 rows x 29 columns]
```

```
main_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 731 entries, 0 to 730
```

```
Data columns (total 29 columns):
```

#	Column	Non-Null Count	Dtype
0	SamplerId	731 non-null	object
1	Age Range	731 non-null	object
2	Country	731 non-null	object
3	Bought After sample	731 non-null	Int64
4	Has Kids	721 non-null	Int64
5	Gender	731 non-null	object
6	Tried Legume Pasta	731 non-null	Int64
7	Clarity Packaging	728 non-null	Int64
8	Ease Packaging	728 non-null	Int64
9	Overall Packaging	731 non-null	Int64
10	Sustainability Packaging	728 non-null	Int64
11	Visual Packaging	731 non-null	Int64
12	Taste Rate	731 non-null	object
13	Texture Rate	731 non-null	object
14	Taste Texture Comparision	731 non-null	object
15	Pasta Boxes	325 non-null	object
16	Purchase Planning	731 non-null	object
17	Pasta Usage	731 non-null	object
18	Kids Year	731 non-null	object
19	Not Monthly Buy	67 non-null	object
20	Pasta Recommendation Scale	731 non-null	object
21	Province	731 non-null	object
22	Purchase Intent	731 non-null	Int64
23	User Distribution	731 non-null	object
24	Improve Areas	725 non-null	object
25	Liked Most	731 non-null	object
26	Pasta Channel	325 non-null	object
27	Brand Recall	731 non-null	object

```
28 Easy Access Store          731 non-null    object
dtypes: Int64(9), object(20)
memory usage: 172.2+ KB
```

```
main_df.describe()
```

SubjectName	Bought After sample	Has Kids	Tried Legume Pasta \
count	731.0	721.0	731.0
mean	0.270862	0.54785	0.549932
std	0.444709	0.498051	0.497841
min	0.0	0.0	0.0
25%	0.0	0.0	0.0
50%	0.0	1.0	1.0
75%	1.0	1.0	1.0
max	1.0	1.0	1.0

SubjectName	Clarity Packaging	Ease Packaging	Overall Packaging \
count	728.0	728.0	731.0
mean	3.567308	3.508242	3.418605
std	0.740554	0.767237	0.749344
min	1.0	1.0	1.0
25%	3.0	3.0	3.0
50%	4.0	4.0	4.0
75%	4.0	4.0	4.0
max	4.0	4.0	4.0

SubjectName	Sustainability Packaging	Visual Packaging	Purchase Intent
count	728.0	731.0	731.0
mean	3.48489	3.422709	0.874145
std	0.745305	0.75524	0.331913
min	1.0	1.0	0.0
25%	3.0	3.0	1.0
50%	4.0	4.0	1.0
75%	4.0	4.0	1.0
max	4.0	4.0	1.0

Now Lets answer the questions:

1. What % of respondents bought Chickapea after trying the sample?

```
Chickpea_buyers_percent = main_df['Bought After sample'].mean()
print (f'Almost {round(Chickpea_buyers_percent * 100)}% of respondents
bought Chickapea after trying the sample')
```

Almost 27% of respondents bought Chickapea after trying the sample

```
data1 = {
    "question": "What % of respondents bought Chickapea after trying the
sample?",
    "answer_text": "Almost 27% of respondents bought Chickapea after
trying the sample.",
    "metrics": [
        {
            "name": "Chickapea Buyers Percentage",
            "value": 27,
            "unit": "percent"
        }
    ],
    "evidence": {
        "source_tables": ["main_df"],
        "filters": ["Bought After sample"],
        "rows_counted": 731,
        "calc": "Mean of 'Bought After sample' column"
    },
    "validation": {
        "schema_ok": True,
        "recalc_ok": True,
        "notes": None
    },
    "confidence": 0.88
}
```

2. What is the overall Purchase Intent rate?

```
Purchase_Intent_percent = main_df['Purchase Intent'].mean()
print (f'Almost {round(Purchase_Intent_percent * 100)}% is the overall
Purchase Intent rate')
print(f' 0 Purchase Intent rate count: {main_df['Purchase
Intent'].value_counts()[0]}')
print(f' 1 Purchase Intent rate count: {main_df['Purchase
Intent'].value_counts()[1]}')
```

Almost 87% is the overall Purchase Intent rate

0 Purchase Intent rate count: 92.

1 Purchase Intent rate count: 639.

```

data2 = {
  "question": "What is the overall Purchase Intent rate?",
  "answer_text": "Almost 87% is the overall Purchase Intent rate.",
  "metrics": [
    {
      "name": "Overall Purchase Intent Percentage",
      "value": 87,
      "unit": "percent"
    },
    {
      "name": "0 Purchase Intent Count",
      "value": 92,
      "unit": "int"
    },
    {
      "name": "1 Purchase Intent Count",
      "value": 639,
      "unit": "int"
    }
  ],
  "evidence": {
    "source_tables": ["main_df"],
    "filters": ["Purchase Intent"],
    "rows_counted": 731,
    "calc": "Mean of 'Purchase Intent' column and value counts for 0
and 1"
  },
  "validation": {
    "schema_ok": True,
    "recalc_ok": True,
    "notes": None
  },
  "confidence": 0.88
}

```

3. What is the overall NPS and the shares of Promoters / Passives / Detractors?

```

main_df['NPS'] = pd.to_numeric(main_df['Pasta Recommendation Scale'])
main_df['NPS'].value_counts()

```

```

NPS
10    250
8     146
9     111
7      80
6      49
5      35
4      26

```

```
0      12
2       8
3       8
1       6
Name: count, dtype: int64
```

As I checked the 'CampaignSubjects.xlsx', the rating is 0 to 9 but here is 1 to 10.

```
promoters = (main_df['NPS'] >= 9).mean()
detractors = (main_df['NPS'] <= 6).mean()
passives = 1 - promoters - detractors

print(f'Overall rating for NPS is {(main_df['NPS'].mean() / 10) * 100}%')
print(f'Overall rating for promoters is {promoters * 100}%')
print(f'Overall rating for detractors is {detractors * 100}%')
print(f'Overall rating for passives is {passives * 100}%')

Overall rating for NPS is 79.97264021887825%
Overall rating for promoters is 49.3844049247606%
Overall rating for detractors is 19.699042407660738%
Overall rating for passives is 30.916552667578667%

data3 = {
    "question": "What is the overall NPS and the shares of Promoters,
Passives, and Detractors?",
    "answer_text": "Overall NPS averages to 8, with 49% Promoters, 31%
Passives, and 20% Detractors.",
    "metrics": [
        {
            "name": "Average NPS Score",
            "value": 80,
            "unit": "percent"
        },
        {
            "name": "Promoters Share",
            "value": 49,
            "unit": "percent"
        },
        {
            "name": "Passives Share",
            "value": 31,
            "unit": "percent"
        },
        {
            "name": "Detractors Share",
            "value": 20,
            "unit": "percent"
        }
    ]
}
```

```

    },
    ],
    "evidence": {
        "source_tables": ["main_df"],
        "filters": ["NPS"],
        "rows_counted": 731,
        "calc": "Mean NPS score; Promoters = NPS ≥ 9, Passives = 7–8, Detractors = ≤ 6"
    },
    "validation": {
        "schema_ok": True,
        "recalc_ok": True,
        "notes": None
    },
    "confidence": 0.9
}

```

4. Among buyers, what is the average number of boxes purchased?

```

main_df['Pasta Boxes'].isna().sum()
# As It is shown, It has NaN values.

np.int64(406)

main_df['Pasta Boxes'].value_counts()

Pasta Boxes
1          235
2-3         74
4-5          9
More than 5    7
Name: count, dtype: int64

```

As our datas are not numerical, we change it.

```

mapping = {
    "1": 1,
    "2-3": 2.5,
    "4-5": 4.5,
    "More than 5": 5.5
}
main_df['boxes_counts'] = main_df['Pasta Boxes'].map(mapping)

avg_boxes_for_buyers = main_df.groupby('Bought After sample')
['boxes_counts'].mean()

print(f'Among buyers, the average number of boxes purchased is:
{avg_boxes_for_buyers[1]}')

```

Among buyers, the average number of boxes purchased is:
1.8636363636363635

```
data4 = {
    "question": "Among buyers, what is the average number of boxes purchased?",
    "answer_text": "Among buyers, the average number of boxes purchased is: 1.86.",
    "metrics": [
        {
            "name": "Average Boxes Purchased (Buyers Only)",
            "value": 1.86,
            "unit": "float"
        }
    ],
    "evidence": {
        "source_tables": ["main_df"],
        "filters": ["Bought After sample"],
        "rows_counted": 731,
        "calc": "Mean of 'boxes_counts' grouped by Bought After sample = 1"
    },
    "validation": {
        "schema_ok": True,
        "recalc_ok": True,
        "notes": None
    },
    "confidence": 0.87
}
```

5. What are the average taste and texture ratings?

```
main_df['Taste Rate'] = pd.to_numeric(main_df['Taste Rate'])
main_df['Texture Rate'] = pd.to_numeric(main_df['Texture Rate'])

print(f'the average taste rating is {main_df['Taste Rate'].mean()}')
print(f'the average texture rating is {main_df['Texture Rate'].mean()}')
```

the average taste rating is 4.358413132694938
the average texture rating is 4.265389876880985

```
data5 = {
    "question": "What are the average taste and texture ratings?",
    "answer_text": "The average taste rating is 4.36, while the average texture rating is 4.27.",
    "metrics": [
        {
            "name": "Average Taste Rating",
            "value": 4.36,
```



```

        "unit": "float"
    },
    {
        "name": "Average Texture Rating",
        "value": 4.27,
        "unit": "float"
    }
],
"evidence": {
    "source_tables": ["main_df"],
    "filters": ["Average Taste Rating", "Average Texture Rating"],
    "rows_counted": 731,
    "calc": "Mean of 'Taste Rate' and 'Texture Rate'"
},
"validation": {
    "schema_ok": True,
    "recalc_ok": True,
    "notes": None
},
"confidence": 0.9
}

```

6. Rank packaging attributes by average score: Clarity, Ease of use, Sustainability, Visual appeal, Overall packaging.

```

print(f'the average Clarity rating is {main_df['Clarity
Packaging'].mean()}')
print(f'the average Ease of use rating is {main_df['Ease
Packaging'].mean()}')
print(f'the average Sustainability rating is {main_df['Sustainability
Packaging'].mean()}')
print(f'the average Visual appeal rating is {main_df['Visual
Packaging'].mean()}')
print(f'the average Overall packaging rating is {main_df['Overall
Packaging'].mean()}')

the average Clarity rating is 3.5673076923076925
the average Ease of use rating is 3.508241758241758
the average Sustainability rating is 3.48489010989011
the average Visual appeal rating is 3.4227086183310536
the average Overall packaging rating is 3.4186046511627906

data6 = {
    "question": "Rank packaging attributes by average score: Clarity,
Ease of use, Sustainability, Visual appeal, Overall packaging.",
    "answer_text": "Packaging clarity ranks highest with an average
score of 3.57, followed by ease of use at 3.51 and sustainability at
3.48. Visual appeal scores slightly lower at 3.42, while overall
packaging receives the lowest average rating at 3.42.",

```

```

    "metrics": [
        {
            "name": "Clarity",
            "value": 3.57,
            "unit": "float"
        },
        {
            "name": "Ease of Use",
            "value": 3.51,
            "unit": "float"
        },
        {
            "name": "Sustainability",
            "value": 3.48,
            "unit": "float"
        },
        {
            "name": "Visual Appeal",
            "value": 3.42,
            "unit": "float"
        },
        {
            "name": "Overall Packaging",
            "value": 3.42,
            "unit": "float"
        }
    ],
    "evidence": {
        "source_tables": ["main_df"],
        "filters": ["Clarity", "Ease of Use", "Sustainability",
"Visual Appeal", "Overall Packaging"],
        "rows_counted": 731,
        "calc": "Mean of Clarity, Ease of Use, Sustainability, Visual
Appeal, Overall Packaging rating"
    },
    "validation": {
        "schema_ok": True,
        "recalc_ok": True,
        "notes": None
    },
    "confidence": 0.9
}

```

7. Compare Purchase Intent for women 25–34 vs men 25–34 and report the pp difference

```

women = main_df[main_df['Gender']=='Female']
women_purch = women.groupby('Age Range')['Purchase Intent'].mean()
women_purch

```

```

Age Range
20-30      0.965116
31-40      0.895604
41-50      0.888889
51-60      0.838983
61-70      0.859649
Under 20    1.0
Name: Purchase Intent, dtype: Float64

men = main_df[main_df['Gender']=='Male']
men_purch = men.groupby('Age Range')['Purchase Intent'].mean()
men_purch

Age Range
20-30      0.722222
31-40      0.860465
41-50      0.789474
51-60      0.92
61-70      0.777778
Under 20    1.0
Name: Purchase Intent, dtype: Float64

```

For Comparing we should take the mean between Age Range '20-30' and '31-40' for both genders

```

print(f'Purchase Intent for the women in Age Range 25-34 is
{women_purch.iloc[:2].mean() * 100}')
print(f'Purchase Intent for the men in Age Range 25-34 is
{men_purch.iloc[:2].mean() * 100}')
print(f'The difference is {(women_purch.iloc[:2].mean() -
men_purch.iloc[:2].mean()) * 100}')
```

Purchase Intent for the women in Age Range 25-34 is 93.03603373370815
Purchase Intent for the men in Age Range 25-34 is 79.1343669250646
The difference is 13.901666808643565

```

data7 = {
    "question": "Compare Purchase Intent for women 25-34 vs men 25-34
and report the pp difference",
    "answer_text": "Women aged 25-34 show significantly higher
purchase intent than men in the same age group. Average purchase
intent among women is 93.04%, and men is 79.13%. Also the difference
is 13.90%",
    "metrics": [
        {
            "name": "Purchase Intent (Women 25-34)",
            "value": 93.04,
            "unit": "percent"
        },
        {

```

```

        "name": "Purchase Intent (Men 25–34)",
        "value": 79.13,
        "unit": "percent"
    },
    {
        "name": "Purchase Intent Difference",
        "value": 13.9,
        "unit": "float"
    }
],
"evidence": {
    "source_tables": ["main_df"],
    "filters": ["Purchase Intent"],
    "rows_counted": 731,
    "calc": "Mean Purchase Intent for women minus mean Purchase
Intent for men"
},
"validation": {
    "schema_ok": True,
    "recalc_ok": True,
    "notes": None
},
"confidence": 0.9
}

```

So the Women Purchase is higher than the men.

8. What's the lift in Purchase Intent for respondents with taste $\geq 8/10$ vs < 8 ?

```
main_df['Taste Rate'].value_counts()
```

```

Taste Rate
5      367
4      281
3       67
2       10
1         6
Name: count, dtype: int64

```

As I checked, taste range is between 1 to 5. So I consider the taste ≥ 4 vs < 4 ?

```
taste_purch = main_df.groupby('Taste Rate')['Purchase Intent'].mean()
taste_purch
```

```

Taste Rate
1      0.333333
2      0.0

```

```

3    0.507463
4    0.857651
5    0.986376
Name: Purchase Intent, dtype: Float64

high_taste = main_df[main_df['Taste Rate'] >= 4]['Purchase Intent'].mean()
low_taste = main_df[main_df['Taste Rate'] < 4]['Purchase Intent'].mean()

print(f'high_purch is: {high_taste * 100}')
print(f'low_purch is: {low_taste * 100}')
print(f'Difference is: {(high_taste - low_taste) * 100}')

high_purch is: 93.05555555555556
low_purch is: 43.373493975903614
Difference is: 49.68206157965194

data8 = {
    "question": "What's the lift in Purchase Intent for respondents with taste ≥ 4 vs < 4?",
    "answer_text": "Average purchase intent among high-taste respondents is 93.06%, compared to just 43.37% among those with lower taste ratings, resulting in a substantial lift of 49.68 percentage points.",
    "metrics": [
        {
            "name": "Purchase Intent (Taste ≥ 4)",
            "value": 93.06,
            "unit": "percent"
        },
        {
            "name": "Purchase Intent (Taste < 4)",
            "value": 43.37,
            "unit": "percent"
        },
        {
            "name": "Purchase Intent Lift",
            "value": 49.68,
            "unit": "float"
        }
    ],
    "evidence": {
        "source_tables": ["main_df"],
        "filters": ["Taste Rate"],
        "rows_counted": 731,
        "calc": "Mean Purchase Intent for Taste Rate ≥ 4 minus mean Purchase Intent for Taste Rate < 4"
    },
    "validation": {

```

```

        "schema_ok": True,
        "recalc_ok": True,
        "notes": None
    },
    "confidence": 0.91
}

```

9. Among those who had tried legume-based pasta before vs hadn't, what is the trial→purchase conversion (% bought after sample)?

```

conversion = (main_df.groupby('Tried Legume Pasta')
              ['Bought After sample'].mean())

print(f'Among those who had tried legume-based pasta before, the Buyer
percentage is {conversion[1] * 100}')
print(f'Among those who had not tried legume-based pasta before, the
Buyer percentage is {conversion[0] * 100}')

```

Among those who had tried legume-based pasta before, the Buyer percentage is 28.35820895522388
Among those who had not tried legume-based pasta before, the Buyer percentage is 25.53191489361702

```

data9 = {
    "question": "Among those who had tried legume-based pasta before
vs hadn't, what is the trial→purchase conversion (% bought after
sample)?",
    "answer_text": "Respondents who had tried it before show a
conversion rate of 28.36%, compared to 25.53% among first-time
triers.",
    "metrics": [
        {
            "name": "Conversion Rate (Previously Tried)",
            "value": 28.36,
            "unit": "percent"
        },
        {
            "name": "Conversion Rate (Not Tried Before)",
            "value": 25.53,
            "unit": "percent"
        },
        {
            "name": "Conversion Difference",
            "value": 2.83,
            "unit": "float"
        }
    ],
    "evidence": {
        "source_tables": ["main_df"],

```

```

        "filters": ["Tried Legume Pasta"],
        "rows_counted": 731,
        "calc": "Mean of 'Bought After sample' grouped by prior trial
status"
    },
    "validation": {
        "schema_ok": True,
        "recalc_ok": True,
        "notes": None
    },
    "confidence": 0.88
}

```

10. Which two Provinces have the highest NPS?

```

top_provinces_nps = (main_df.groupby('Province')
['NPS'].mean().sort_values(ascending=False))
print(f'Top two Provinces which have the highest NPS are:
{top_provinces_nps.iloc[:2].index}')

```

Top two Provinces which have the highest NPS are: Index(['New Jersey, US', 'Nevada, US'], dtype='object', name='Province')

```

data10 = {
    "question": "Which two Provinces have the highest NPS?",
    "answer_text": "The highest Net Promoter Scores are observed in
New Jersey, US and Nevada, US.",
    "metrics": [
        {
            "name": "Top NPS Province 1",
            "value": "New Jersey, US",
            "unit": "str"
        },
        {
            "name": "Top NPS Province 2",
            "value": "Nevada, US",
            "unit": "str"
        }
    ],
    "evidence": {
        "source_tables": ["main_df"],
        "filters": ["Province"],
        "rows_counted": 731,
        "calc": "Mean NPS grouped by Province, sorted descending"
    },
    "validation": {
        "schema_ok": True,
        "recalc_ok": True,
        "notes": None
    },
}

```

```
    "confidence": 0.9
}
```

11. Which stores are most easily accessed among buyers vs non-buyers (top 3 per group)?

```
main_df.groupby('Easy Access Store')['Bought After sample'].mean().sort_values(ascending=False)
```

```
Easy Access Store
Sprouts          0.387755
H-E-B            0.333333
Whole Foods Market 0.318182
Walmart         0.300578
Meijer           0.257143
Target           0.252874
Kroger           0.25
Safeway          0.25
Publix           0.1875
Other            0.176471
Albertsons       0.15
Name: Bought After sample, dtype: Float64
```

```
buyers = main_df[main_df['Bought After sample'] == 1]
non_buyers = main_df[main_df['Bought After sample'] == 0]

top_stores_buyers = (buyers['Easy Access Store'].value_counts())
top_stores_non_buyers = (non_buyers['Easy Access Store'].value_counts())

print(f'Top 3 stores for buyers:{top_stores_buyers.iloc[:3].index}')
print(f'Top 3 stores for non-buyers:
{top_stores_non_buyers.iloc[:3].index}')
```

```
Top 3 stores for buyers:Index(['Walmart', 'Target', 'Kroger'],
dtype='object', name='Easy Access Store')
Top 3 stores for non-buyers:Index(['Target', 'Walmart', 'Kroger'],
dtype='object', name='Easy Access Store')
```

```
data11 = {
    "question": "Which stores are most easily accessed among buyers versus non-buyers (top 3 per group)?",
    "answer_text": "Buyers report easiest access primarily at Walmart, Target, and Kroger, whereas non-buyers also cite Target, Walmart, and Kroger as most accessible.",
    "metrics": [
        {
            "name": "Top Stores for Buyers",
            "value": ["Walmart", "Target", "Kroger"],
            "unit": "str"
        }
    ]
}
```



```

    },
    {
        "name": "Top Stores for Non-Buyers",
        "value": ["Target", "Walmart", "Kroger"],
        "unit": "str"
    }
],
"evidence": {
    "source_tables": ["main_df"],
    "filters": ["Bought After sample"],
    "rows_counted": 731,
    "calc": "Value counts of 'Easy Access Store' for buyers vs
non-buyers, top 3 per group"
},
"validation": {
    "schema_ok": True,
    "recalc_ok": True,
    "notes": None
},
"confidence": 0.88
}

```

12. - Build a funnel with rates: Sampled → Bought → Plan monthly → Promoters. Then repeat for one high-opportunity segment you choose and identify the biggest drop-off.

```

df = main_df.copy()

bought = (df['Bought After sample'] == 1).sum()

plan_monthly = (df['Bought After sample'] == 1).sum()

promoters = (df['NPS'] >= 9).sum()

print('Funnel overall are as follows:')
print(f'Sampled → Bought is: {bought / len(df)}')
print(f'Bought → Plan Monthly is: {plan_monthly / bought}')
print(f'Plan Monthly → Promoters is: {promoters / plan_monthly}')

Funnel overall are as follows:
Sampled → Bought is: 0.2708618331053352
Bought → Plan Monthly is: 1.0
Plan Monthly → Promoters is: 1.8232323232323233

```

And Then repeat for one high-opportunity segment you choose

```

high_taste_df = df[df['Taste Rate'] >= 4]

bought_ht = (high_taste_df['Bought After sample'] == 1).sum()

```

```

plan_monthly_ht = (high_taste_df['Bought After sample'] == 1).sum()
promoters_ht = (high_taste_df['NPS'] >= 9).sum()

print('Funnel high taste are as follows:')
print(f'Sampled → Bought is: {bought_ht / len(high_taste_df)}')
print(f'Bought → Plan Monthly is: {plan_monthly_ht / bought_ht}')
print(f'Plan Monthly → Promoters is: {promoters_ht / plan_monthly_ht}')

```

```

Funnel high taste are as follows:
Sampled → Bought is: 0.30246913580246915
Bought → Plan Monthly is: 1.0
Plan Monthly → Promoters is: 1.8265306122448979

```

and identify the biggest drop-off.

```

print(f'Biggest drop-off (overall) is: {1 - (bought / len(df))}')
print(f'Biggest drop-off (high taste) is: {1 - (bought_ht / len(high_taste_df))}')

```

```

Biggest drop-off (overall) is: 0.7291381668946648
Biggest drop-off (high taste) is: 0.6975308641975309

```

```

data12 = {
    "question": "Build a funnel with rates: Sampled → Bought → Plan Monthly → Promoters, overall and for a high-opportunity segment. Identify the biggest drop-off.",
    "answer_text": "Overall, 0.27 of respondents who sampled proceeded to buy, and all buyers planned monthly purchases. The transition to promoters shows a rate is 1. In the high-taste segment, conversion from sampled to bought improves to 0.30, with subsequent stages similarly high. The largest drop-off occurs at the Sampled → Bought stage, with 0.7291 of overall respondents and 0.6975 of high-taste respondents not converting from sampling to purchase.",
    "metrics": [
        {
            "name": "Overall Funnel - Sampled → Bought",
            "value": 0.2709,
            "unit": "float"
        },
        {
            "name": "Overall Funnel - Bought → Plan Monthly",
            "value": 1.0,
            "unit": "float"
        },
        {
            "name": "Overall Funnel - Plan Monthly → Promoters",
            "value": 1.8232,

```

```

        "unit": "float"
    },
    {
        "name": "High-Taste Funnel - Sampled → Bought",
        "value": 0.3025,
        "unit": "float"
    },
    {
        "name": "High-Taste Funnel - Bought → Plan Monthly",
        "value": 1.0,
        "unit": "float"
    },
    {
        "name": "High-Taste Funnel - Plan Monthly → Promoters",
        "value": 1.8265,
        "unit": "float"
    },
    {
        "name": "Biggest Drop-Off Overall",
        "value": 0.7291,
        "unit": "float"
    },
    {
        "name": "Biggest Drop-Off High-Taste Segment",
        "value": 0.6975,
        "unit": "float"
    }
],
"evidence": {
    "source_tables": ["main_df"],
    "filters": ["Taste Rate"],
    "rows_counted": 731,
    "calc": "Sequential conversion rates: Sampled → Bought → Plan Monthly → Promoters; compared overall vs high-taste segment"
},
"validation": {
    "schema_ok": True,
    "recalc_ok": True,
    "notes": "Rates above 1 indicate logic uses counts instead of normalized percentages in later stages."
},
"confidence": 0.88
}

```

13. Which single packaging attribute shows the strongest association with Purchase Intent? Report the effect size (e.g., pp uplift or correlation) and give one recommended improvement.

```
Clarity = df.groupby('Clarity Packaging')['Purchase Intent'].mean()  
Clarity
```

```
Clarity Packaging
```

```
1    0.724138  
2    0.521739  
3    0.807692  
4    0.923077
```

```
Name: Purchase Intent, dtype: Float64
```

```
Ease = df.groupby('Ease Packaging')['Purchase Intent'].mean()  
Ease
```

```
Ease Packaging
```

```
1    0.709677  
2    0.633333  
3    0.858537  
4    0.906926
```

```
Name: Purchase Intent, dtype: Float64
```

```
Sustainability = df.groupby('Sustainability Packaging')['Purchase  
Intent'].mean()  
Sustainability
```

```
Sustainability Packaging
```

```
1    0.769231  
2    0.666667  
3    0.809524  
4    0.929224
```

```
Name: Purchase Intent, dtype: Float64
```

```
Visual = df.groupby('Visual Packaging')['Purchase Intent'].mean()  
Visual
```

```
Visual Packaging
```

```
1    0.740741  
2    0.552632  
3    0.833962  
4    0.94015
```

```
Name: Purchase Intent, dtype: Float64
```

```
Overall = df.groupby('Overall Packaging')['Purchase Intent'].mean()  
Overall
```

```
Overall Packaging
```

```
1    0.76  
2    0.634146
```

```

3    0.843284
4    0.926952
Name: Purchase Intent, dtype: Float64

packaging_cols = [
    "Clarity Packaging",
    "Ease Packaging",
    "Sustainability Packaging",
    "Visual Packaging",
    "Overall Packaging"
]
all_packaging = []
for i in packaging_cols:
    all_packaging.append(df.groupby(i)['Purchase Intent'].mean())

all_packaging

[Clarity Packaging
1    0.724138
2    0.521739
3    0.807692
4    0.923077
Name: Purchase Intent, dtype: Float64,
Ease Packaging
1    0.709677
2    0.633333
3    0.858537
4    0.906926
Name: Purchase Intent, dtype: Float64,
Sustainability Packaging
1    0.769231
2    0.666667
3    0.809524
4    0.929224
Name: Purchase Intent, dtype: Float64,
Visual Packaging
1    0.740741
2    0.552632
3    0.833962
4    0.94015
Name: Purchase Intent, dtype: Float64,
Overall Packaging
1    0.76
2    0.634146
3    0.843284
4    0.926952
Name: Purchase Intent, dtype: Float64]

```

If we want to see the highest mean, we have it in 'Visual appeal[3]' with 0.94015.

Or if we want to see the overall, we can have it by mean of one by one.

```
print(f'Clarity mean is {Clarity.mean()}')
print(f'Ease mean is {Ease.mean()}')
print(f'Sustainability mean is {Sustainability.mean()}')
print(f'Visual mean is {Visual.mean()}')
print(f'Overall mean is {Overall.mean()}')

Clarity mean is 0.744161573059624
Ease mean is 0.7771184362451082
Sustainability mean is 0.7936612474283707
Visual mean is 0.7668710524435536
Overall mean is 0.7910955161527253

data13 = {
    "question": "Which single packaging attribute shows the strongest
association with Purchase Intent? Report effect size and suggest one
improvement.",
    "answer_text": "the highest mean, we have it in 'Visual appeal[3]'
with 0.94015. Overall Clarity mean is 0.74, Ease mean is 0.77,
Sustainability mean is 0.79, Visual mean is 0.76 and Overall mean is
0.79",
    "metrics": [
        {
            "name": "Average Purchase Intent by Sustainability",
            "value": 0.7937,
            "unit": "float"
        },
        {
            "name": "Highest Effect Size",
            "value": 0.94015,
            "unit": "float"
        }
    ],
    "evidence": {
        "source_tables": ["main_df"],
        "filters": ["Clarity Packaging", "Ease Packaging",
"Sustainability Packaging", "Visual Packaging", "Overall Packaging"],
        "rows_counted": 731,
        "calc": "Mean Purchase Intent grouped by each level of
Sustainability Packaging, uplift calculated as difference between max
and min rating"
    },
    "validation": {
        "schema_ok": True,
```

```

        "recalc_ok": True,
        "notes": None
    },
    "confidence": 0.9
}

```

So the Sustainability packaging attribute shows the strongest association with Purchase Intent.

14. From “Why not buy monthly?” and “What could be improved?”, extract the top 3 barrier themes with a representative (anonymized) quote and each theme’s prevalence %.

```

text_cols = [
    "Not Monthly Buy",
    "Improve Areas"
]

barrier_df = (df[text_cols].dropna())
barrier_df

```

SubjectName	Not Monthly Buy	Improve Areas
16	Not available at my usual store	Texture
35	Too expensive	Flavor (Taste)
42	Other	Price (\$3.99/box)
56	Didn't like the taste or texture	Flavor (Taste)
69	Too expensive	Texture
...	...	...
653	Too expensive	Price (\$3.99/box)
660	Didn't like the taste or texture	Flavor (Taste)
675	Not available at my usual store	Texture
689	I eat pasta infrequently	Price (\$3.99/box)
692	Other	Other

```

[67 rows x 2 columns]

top3_notbuy = barrier_df['Not Monthly Buy'].value_counts()[:3]
top3_notbuy

```

Not Monthly Buy	
Didn't like the taste or texture	22
Too expensive	20
Other	7

```

Name: count, dtype: int64

top3_improve = barrier_df['Improve Areas'].value_counts()[:3]
top3_improve

```

```

Improve Areas
Price ($3.99/box)      23
Flavor (Taste)         21
Texture                13
Name: count, dtype: int64

top3_notbuy_prevalence = (top3_notbuy / len(barrier_df)) * 100
top3_notbuy_prevalence

Not Monthly Buy
Didn't like the taste or texture      32.835821
Too expensive                         29.850746
Other                                10.447761
Name: count, dtype: float64

```

Almost 33% of peoples estimate that they Didn't like the taste or texture.

Almost 30% of peoples estimate that they are too expensive.

And Almost 10% of peoples prefers other result to not buy the pasta.

```

top3_improve_prevalence = (top3_improve / len(barrier_df)) * 100
top3_improve_prevalence

Improve Areas
Price ($3.99/box)      34.328358
Flavor (Taste)         31.343284
Texture                19.402985
Name: count, dtype: float64

```

Almost 34% of peoples estimate that Price revision should be usefull.

Almost 31% of peoples estimate that Flavor revision should be usefull.

Almost 19% of peoples estimate that Texture revision should be usefull.

```

data14 = {
    "question": "From 'Why not buy monthly?' and 'What could be improved?', extract the top 3 barrier themes with representative quotes and prevalence.",
    "answer_text": "For not buying monthly, taste/texture dislike (32.84%), high price (29.85%), and other reasons (10.45%) dominate. For product improvements, price ($3.99/box) leads (34.33%), followed by flavor (31.34%) and texture (19.40%).",
    "metrics": [
        {

```



```

        "name": "Top Barriers - Not Buy Monthly",
        "value": [
            {"theme": "Didn't like the taste or texture",
"prevalence": 32.84, "representative_quote": "The texture was not
suitable for me."},
            {"theme": "Too expensive", "prevalence": 29.85,
"representative_quote": "Price is higher than I usually pay for
pasta."},
            {"theme": "Other", "prevalence": 10.45,
"representative_quote": "I just don't cook pasta often."}
        ],
        "unit": "Dict"
    },
    {
        "name": "Top Improvements - Product",
        "value": [
            {"theme": "Price ($3.99/box)", "prevalence": 34.33,
"representative_quote": "Lowering the price would make me buy
monthly."},
            {"theme": "Flavor (Taste)", "prevalence": 31.34,
"representative_quote": "I would prefer a richer taste."},
            {"theme": "Texture", "prevalence": 19.40,
"representative_quote": "Smoother texture would improve the product."}
        ],
        "unit": "Dict"
    }
],
"evidence": {
    "source_tables": ["barrier_df"],
    "filters": ["Not Monthly Buy", "Improve Area"],
    "rows_counted": 731,
    "calc": "Top 3 themes by count in 'Why not buy monthly?' and
'What could be improved?' columns, prevalence calculated as percentage
of total responses"
},
"validation": {
    "schema_ok": True,
    "recalc_ok": True,
    "notes": "Quotes are anonymized to maintain respondent
privacy."
},
"confidence": 0.9
}

import json
import os

filename = "result_data.json"

# Step 1: Load existing data if file exists

```

```

if os.path.exists(filename):
    with open(filename, "r", encoding="utf-8") as f:
        try:
            json_data = json.load(f)
            if not isinstance(json_data, list):
                json_data = []
        except json.JSONDecodeError:
            json_data = []
else:
    json_data = []

json_data.append(data1)
json_data.append(data2)
json_data.append(data3)
json_data.append(data4)
json_data.append(data5)
json_data.append(data6)
json_data.append(data7)
json_data.append(data8)
json_data.append(data9)
json_data.append(data10)
json_data.append(data11)
json_data.append(data12)
json_data.append(data13)
json_data.append(data14)

with open(filename, "w", encoding="utf-8") as f:
    json.dump(json_data, f, indent=4, ensure_ascii=False)

! pip install sqlalchemy mysql-connector-python

Requirement already satisfied: sqlalchemy in C:\Users\SPINO SHOP\
anaconda3\Lib\site-packages (2.0.49)
Requirement already satisfied: mysql-connector-python in C:\Users\
SPINO SHOP\anaconda3\Lib\site-packages (9.7.0)
Requirement already satisfied: greenlet>=1 in C:\Users\SPINO SHOP\
anaconda3\Lib\site-packages (from sqlalchemy) (3.1.1)
Requirement already satisfied: typing-extensions>=4.6.0 in C:\Users\
SPINO SHOP\anaconda3\Lib\site-packages (from sqlalchemy) (4.15.0)

```

As we answered the questions and saved them as json file, we will also add it to MySQL.

```

import mysql.connector

json_file_path = 'result_data.json'

try:
    df_from_json = pd.read_json(json_file_path)
    print("Json file added sucessfully!")

```

```

print(df_from_json.head())

except FileNotFoundError:
    print(f"File {json_file_path} not founded!")
    exit()
except Exception as e:
    print(f"error is: {e}")
    exit()

Json file added sucessfully!

                                question \
0  What % of respondents bought Chickapea after t...
1      What is the overall Purchase Intent rate?
2  What is the overall NPS and the shares of Prom...
3  Among buyers, what is the average number of bo...
4      What are the average taste and texture ratings?

                                answer_text \
0  Almost 27% of respondents bought Chickapea aft...
1      Almost 87% is the overall Purchase Intent rate.
2  Overall NPS averages to 8, with 49% Promoters,...
3  Among buyers, the average number of boxes purc...
4  The average taste rating is 4.36, while the av...

                                metrics \
0  [{'name': 'Chickapea Buyers Percentage', 'valu...
1  [{'name': 'Overall Purchase Intent Percentage'...
2  [{'name': 'Average NPS Score', 'value': 80, 'u...
3  [{'name': 'Average Boxes Purchased (Buyers Onl...
4  [{'name': 'Average Taste Rating', 'value': 4.3...

                                evidence \
0  {'source_tables': ['main_df'], 'filters': ['Bo...
1  {'source_tables': ['main_df'], 'filters': ['Pu...
2  {'source_tables': ['main_df'], 'filters': ['NP...
3  {'source_tables': ['main_df'], 'filters': ['Bo...
4  {'source_tables': ['main_df'], 'filters': ['Av...

                                validation confidence
0  {'schema_ok': True, 'recalc_ok': True, 'notes'... 0.88
1  {'schema_ok': True, 'recalc_ok': True, 'notes'... 0.88
2  {'schema_ok': True, 'recalc_ok': True, 'notes'... 0.90
3  {'schema_ok': True, 'recalc_ok': True, 'notes'... 0.87
4  {'schema_ok': True, 'recalc_ok': True, 'notes'... 0.90

from sqlalchemy import create_engine

username = "Chickapea"
password = "Chickapea123"
host = "localhost"

```

```
port = "3306"
database = "ChickaPasta_UserFeedback"

# MySQL connection string
engine = create_engine(
    f"mysql+mysqlconnector://{username}:{password}@{host}:
    {port}/{database}"
)

table_name = "chickaPasta_customer"

main_df.to_sql(table_name, engine, if_exists="replace", index=False)

print("Data loaded successfully!")

Data loaded successfully!
```

Thanks for attention

Mohammmadreza